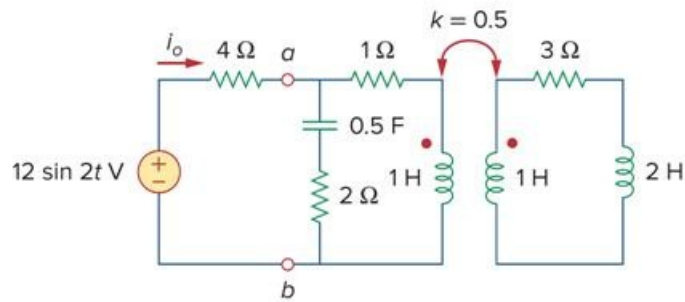


Problem

For the network in Fig. 13.94, find $\mathbf{Z}_{ab}$ and $\mathbf{I}_o$.

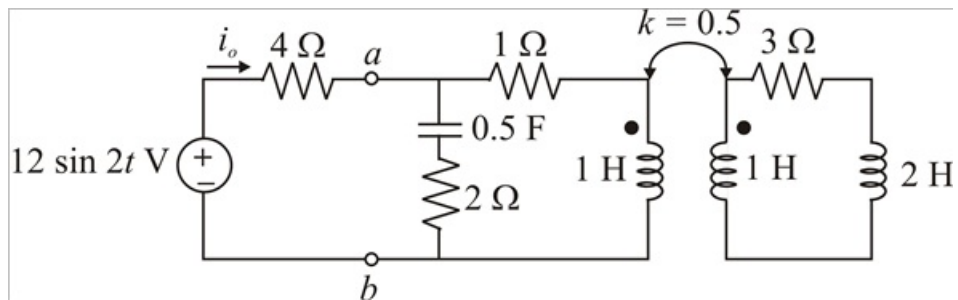
Figure 13.94



Step-by-step solution

Step 1 of 10

Given circuit diagram is:



Step 2 of 10

We first convert the time domain circuit to frequency domain circuit as shown below:

$$12 \sin 2t \text{ V} \Rightarrow 12 \cos(2t - 90^\circ) \quad \left(\because \mp \sin x = \cos(x \pm 90^\circ) \right)$$

$$12 \sin 2t \text{ V} \Rightarrow 12 \angle -90^\circ \text{ V.}$$

Here, we have

$$\omega = 2 \text{ rad/s}$$

The coupling coefficient is

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

$$M = k \sqrt{L_1 L_2}$$

$$M = 0.5 \sqrt{1 \times 1}$$

$$M = 0.5 \text{ H}$$

Step 3 of 10

$$X_M = j\omega M$$

$$X_{0.5H} = j \times 2 \times 0.5$$

$$X_{0.5H} = j1\Omega$$

$$X_L = j\omega L$$

$$X_{1H} = j \times 2 \times 1$$

$$X_{1H} = j2\Omega$$

$$X_L = j\omega L$$

$$X_{2H} = j \times 2 \times 2$$

$$X_{2H} = j4\Omega$$

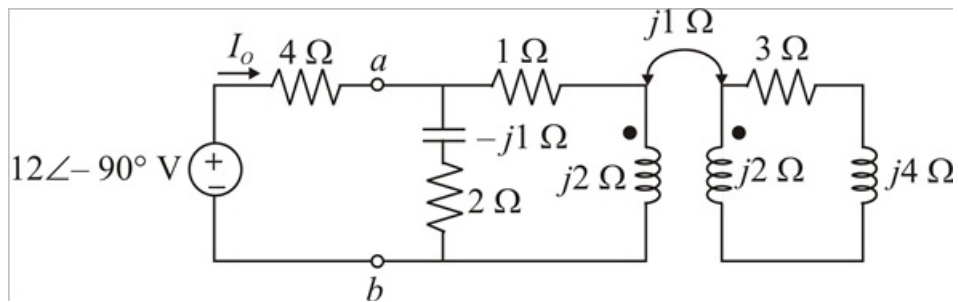
$$X_C = \frac{1}{j\omega C}$$

$$X_{0.5F} = \frac{1}{j \times 2 \times 0.5}$$

$$X_{0.5F} = -j1\Omega$$

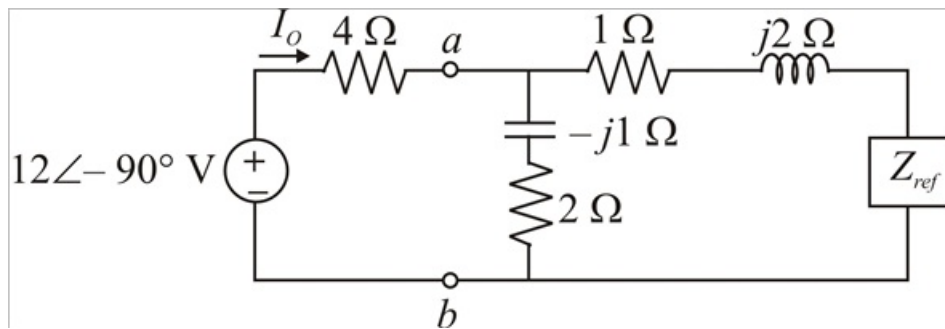
Thus, the frequency domain equivalent circuit is

Step 4 of 10



Step 5 of 10

Using *reflected impedance* concept, above circuit can be modified as below



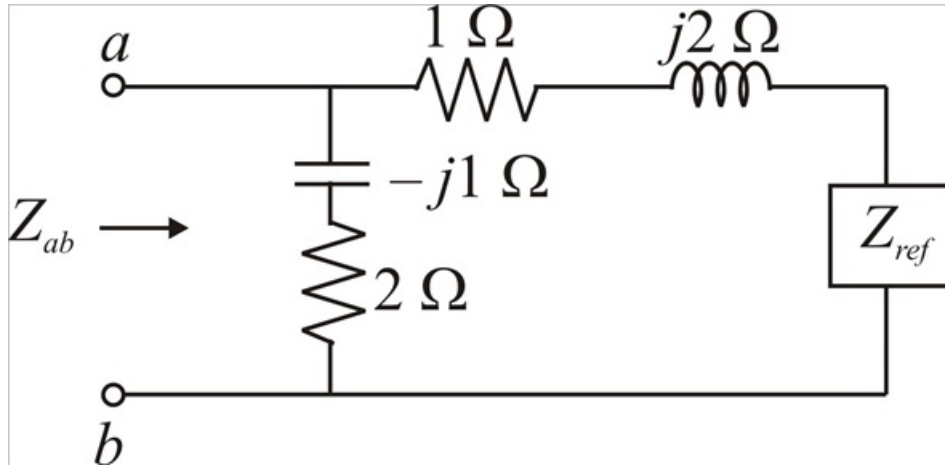
Step 6 of 10

Where,

$$\begin{aligned}
 Z_{ref} &= \frac{(1)^2}{j2 + 3 + j4} \\
 &= \frac{1}{3 + j6} \\
 &= \frac{1}{3 + j6} \times \frac{3 - j6}{3 - j6} \\
 &= \frac{3 - j6}{3^2 + 6^2} \quad (\because j^2 = -1) \\
 &= \frac{3 - j6}{45}
 \end{aligned}$$

$$Z_{ref} = 0.0667 - j0.1333 \Omega$$

For finding Z_{ab} , consider the below circuit



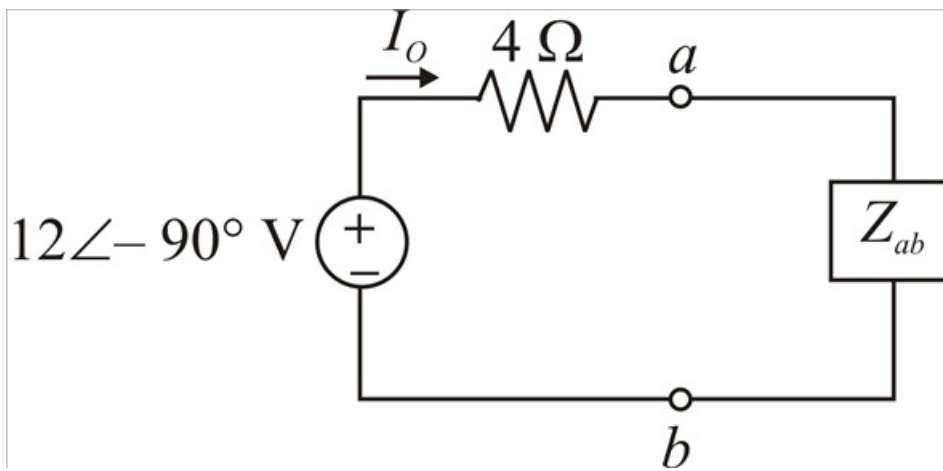
Step 7 of 10

From the above circuit,

$$\begin{aligned}
 Z_{ab} &= (2 - j1) \parallel (1 + j2 + Z_{ref}) \\
 &= (2 - j1) \parallel (1 + j2 + 0.0667 - j0.1333 \Omega) \\
 &= (2 - j1) \parallel (1.0667 + j1.8667) \\
 &= \frac{(2 - j1) \times (1.0667 + j1.8667)}{(2 - j1) + (1.0667 + j1.8667)} \\
 &= \frac{4 + j2.667}{3.0667 + j0.8667} \\
 &= \frac{4.8076 \angle 33.69^\circ}{3.187 \angle 15.78^\circ} \\
 \boxed{Z_{ab} = 1.5085 \angle 17.9^\circ \Omega}
 \end{aligned}$$

Step 8 of 10

Given circuit can be represented as below:



Step 9 of 10

Using Ohm's law to the above circuit

$$\begin{aligned} I_o &= \frac{12\angle -90^\circ}{4 + Z_{ab}} \\ &= \frac{12\angle -90^\circ}{4 + 1.5085\angle 17.9^\circ} \\ &= \frac{12\angle -90^\circ}{4 + (1.435 + j0.464)} \\ &= \frac{12\angle -90^\circ}{5.435 + j0.464} \\ &= \frac{12\angle -90^\circ}{5.455\angle 4.88^\circ} \\ I_o &= 2.2\angle -94.88^\circ \text{ A} \end{aligned}$$

Step 10 of 10

Converting frequency domain current to time domain current, we get

$$\begin{aligned} i_o &= 2.2 \cos(2t - 94.88^\circ) \\ i_o &= 2.2 \sin(2t - 94.88^\circ + 90^\circ) \quad \left(\because \cos(x \pm 90^\circ) = \mp \sin x \right) \\ \boxed{i_o} &= 2.2 \sin(2t - 4.88^\circ) \text{ A} . \end{aligned}$$